Research Statement

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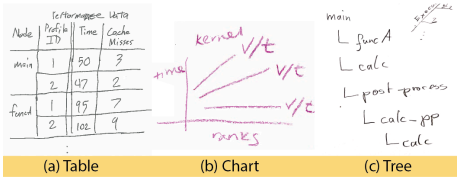
1. Introduction

Visualization research has a subject problem. Current work exploring data analysts and analytical tools tends to focus on a fairly specific type of user: a highly data literate, expert analyst or "data scientist." This focus has historically excluded a voluminous group frequently working with data: "data workers" [1], [2], [3] These data workers–defined as people who use data as part of their broader professional duties but do not identify as data professionals–are the focus of my research and exhibit many unique characteristics as analysts that drive my past and future research goals. These individuals workflows are inherently federated across sets of commercial software (e.g. Excel, Tableau, ArcGIS) to execute increasingly essential data analysis tasks relevant to their professional duties. This population lives on the periphery of frontier research frequently using custom made and generic software that works "just good enough" for their needs. This group survives in a local minima and suffers real consequences for inefficient workflows. Frequent context swapping with little direct integration between tools lead to *substantial productivity losses@*, *missed insights* as users abandon desired analytical tasks which no tool supports especially well[1], and even *critical failures* when data interfaces cannot synchronize relevant data across tools fast enough (e.g. Southwest Airlines' 2022 mass flight cancellation). Currently, LLMs stand to impact these individuals significantly as AI integration builds among commercial software platforms but also poses risks with uneven and inconsistent adoption among federated workflows. These workers and the issues they experience with their workflows inform the primary research gaps I investigate. First, data interface designers lack explicit guidance on effective design methods for building tools that merge traditionally federated workflows and reduce the impact of context swaps. Second, domain specific data workers require more effective, flexible tools that adhere to their mental data models for executing exploratory and confirmatory analysis with their data.

To span these research gaps, **I design, build, and evaluate data science tools for domain-specific data workers.** I accomplish this via three interconnected research and development activities: 1. I use qualitative and ethnographic research methods to characterize how experts reason about and work with data; 2. I translate user characterizations into design decisions about where functionality should live in multimodal analysis tools and communicate these decisions as models for re-use by the design community; and, 3. I use software engineering techniques to build and deploy reliable, usable software that endure and provide a foundation for future longitudinal work. My work has been published in IEEE Transactions on Visualization and Computer Graphics (TVCG), ACM International Symposium on High-Performance Parallel and Distributed Computing (HPDC), and was awarded first prize in the ACM Student Research Competition at the IEEE/ACM International Conference for High Performance Computing, Networking, Storage and Analysis (SC). My software has been deployed and used three national labs across the country. My ongoing and future work explores the technical, cognitive, and cultural gaps that impact the adoption and design of analytical tools in federated workflows.

2. Multimodal Approaches to Data Interaction

Scripting-based environments called computational notebooks(cite) have become the dominant platform for analytical work undertaken by data scientists and some data workers yet domain-specific tools that meaningfully span data interaction modalities (scripting and visualization) are uncommon. In High Performance Computing in particular, data visualizations are mostly built up from chart authoring libraries like Matplotlib(cite) and Seaborn(cite) and act as analytical endpoints where data goes in and does not come back out. Analysts must manually re-construct transformations and queries made in a point a click interface to push forward with contextual insights. More complex visualizations and data exploration tools are often only available as separate software suites [4] – inhibiting smooth transitions from visual investigations to script-based analytics. Identifying that gaps like these are grounded in a lack of design guidance around how to divide functional responsibilities between visualization or scripting, **I use ethnographic qualitative research techniques to deeply understand data scientist's and data worker's (cite data workers) mental models of their data** and **develop guidelines for designing data interaction tools that span visual and script-based representations.**



(a) Table (b) Chart (c) Tree

Figure 1: Examples of drawn data representations produced by participants.

As a foundational step in the non-trivial work of modeling cross-representation user interactions in notebooks; I engaged in a series of interviews with ten data workers about a complex dataset with which they were all deeply familiar[5]. To capture as many facets of these participants' conceptions of their dataset we used multiple different elicitation techniques when interviewing them. Participants were asked to describe, draw (Figure 1) and envision working with their data. Using a multi-phase reflexive thematic analysis over these qualitative datasets, I observed a surprising diversity of data models evidenced by a relatively close analytical team describing one *shared dataset* with a *pre-existing*

canonical model. Participants listed different conceptual data structures and prioritized different components of the data. This diversity highlights the necessity of equally diverse and flexible interfaces for data exploration as the "important" elements of a dataset can be personal and subjective. My subsequent design work builds upon this work by exploring *how* we can manage this flexibility through multimodal tool design.

Traditional design guidelines and research are often focused on one modality of data representation (e.g. visualization, scripting, spreadsheet etc.). This gap leaves the would-be multimodal data interface designer with outstanding questions about where and when a particular functionality should be built into a visualization tool, deferred to scripting tools, duplicated in both or supported as one continuous workflow *across* both representations. I explore these fundamental design questions through qualitatively grounded interviews and embedded research methodologies as a part of two design studies. In one study, I developed a model for assigning representational responsibility based around specific characteristics of abstracted user tasks–specifically specificity and frequency of a task. Building upon expert interviews and surveys of common analysis tasks undertaken by data analysts in HPC performance engineering space, this conceptual framework (Figure 2) structures highly complex multimodal design problems by categorizing more frequent and less specific tasks (e.g. dataset exploration) as well suited to visualization while more specific and less frequent tasks (e.g. query building, model development) are often better fit by scripting. Across two evaluations of the visualization we build using this model, we found that this design helped serve expert users needs by providing them with all the functionality they required from point-and-click software while allowing them to drop-out of the visualization back to their familiar scripting tools. This work was published in IEEE Transactions on Visualization and Computer Graphics [6].

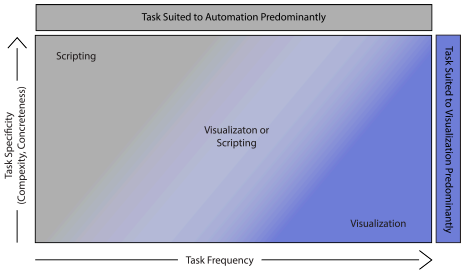


Figure 2: Model for assigning tasks to visualization or scripting when designing an analysis tool.

In a follow-on design project, my task-based division proved insufficient for serving user needs as many different stakeholder groups expressed interest in the one data analysis tool being developed. A simple task list from many different groups obscured the potential impact and goals served by implementing a particular functionality in a visualization. Even if the task was well suited to a visualization according to its frequency and specificity, it may be unworthy of the development effort if the impact to specific populations of users is unclear. To resolve this issue of weighing implementation choices, I cast user groups into well defined *personas*, following guidance from persona-characterizing literature(cite) and using data collected from eight prototype feedback interviews with diverse expert users. With these personas, I identified what tasks were shared across groups and which were unique to each group (Figure 3). The resulting multimodal design of a tool in this instance can then be identified based on the overlap between personas; with shared tasks warranting support by a visual tool and disjoint tasks being left to existing scripting workflows. For example, even though one set of users may desire robust modeling capabilities built into a visual tool, if no other groups need them, then such a tool need not support modeling but can provide a data structure containing potential predicates of interest which may help with building a model computationally. Although this provides a promising model for dividing up responsibilities among scripting and visualization it poses further questions about what support we can continue to provide users after they leave an interactive tool which informs my current work on developing LLM-powered analysis assistants to support confirmatory analysis. This work was published in IEEE Transactions on Visualization and Computer Graphics [7].

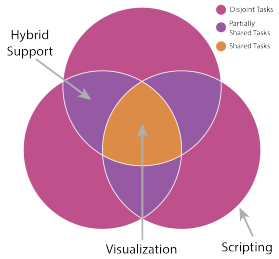


Figure 3: General model for designing for multiple user groups.

3. Research in Practice Through Collaborations with National Labs

Even with the support of multimodal design guidance, domain specific analysis tools–like interactive data visualizations–must also serve the analytical needs of users *in practice* by enabling the investigation of domain-specific data in service of answering complex research questions. As analytical workflows become increasingly interactive and dependant on dynamic, abundant data, traditional tools for descriptive visualization (dashboards, static visual outputs) and generic charting libraries (MatplotLib, Altair) introduce points of friction for users. For example, generic charting libraries often do not meaningfully support hierarchical data that can be common in domains like HPC and also rarely support rendering (or intelligently eliding) the high volume of data which users wish to see. Dashboards and static chart outputs can serve as effective tools for communicating results and critical issues but provide no practical interface for extending analysis from visually-derived insight directly to a code-based confirmatory analysis. To ease these points of friction and aid users in the domain of high performance computing analysis, **I use software engineering and visualization**

design practices to build practical analysis software that is fully deployed in multiple software libraries actively used by analysts across the US National Laboratory System.

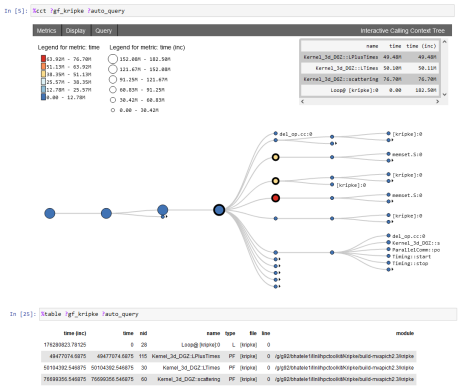


Figure 4: %cct visualization.

Among the many types of HPC-specific data that is currently underserved by analytical tooling, software performance data poses particularly complex visualization and software design problems. This data is collected by measuring the runtime and resource usage of simulation or modeling software running on shared supercomputers. Typically very large and hierarchical (encoding function calls as a tree of calling contexts), the use of generic charting software to plot this data is often impractical. While analysis tooling exists for exploring this data, they mostly exists as standalone platforms removed from code-based analysis workflows (INTEL, NVIDIA NSIGHT, HPCTOOLKIT), incurring costly productivity delays through substantial context swaps (context swap citation). To resolve these issues, I developed **%cct**: an interactive, notebook-embedded visualization designed to support the rendering and querying of specific nodes in very large calling context trees (CCT cite) (Figure 4). The visual design innovates on prior CCT visualizations by using multiple encoding channels to highlight particularly salient points of slowdown and elides noisy function calls which are not likely to be relevant to analysis. Fluent movement between coding and visualization-based analysis is supported through a custom-built library[1] that enabled the automatic syncing of data queried in the visualization back to the notebook context. The **%cct** visualization and functionality has been deployed in two analysis tools: *Hatchet*[8] and *Thicket*[9], used by researchers at Lawrence Livermore National Laboratory and the University of Maryland. Insights from this project were leveraged to enhance another software platform for exploring performance data **Traveler**[10].



Figure 5: Guidepost visualization for supercomputer queue data.

While performance data is a commonly occurring subject of analysis in HPC contexts, there is an even more commonly analyzed and unevenly served type of data generated by HPC systems: jobs submission and telemetry data. Like a bedside monitor in a hospital, this data encapsulates the minute-to-minute operational health of a supercomputer: conveying everything from the wait times users experience for their software to run, to the energy usage of individual GPUs. Although *in situ* visualization of this data is historically robust with dashboards surfacing a variety of problems as they occur; post hoc analysis of this data is significantly less well served by visualization tools tailored to this domain. As this data is tabular, dense, multivariate, and collected at varying time resolutions, significant onus is placed on data analysts to wrangle this data in advance of generating basic visualizations for even initial exploratory analysis. To resolve these issues that inhibit quick exploratory analysis of this data along multiple dimensions I developed **Guidepost**: an interactive multivariate visualization tailored to surface interesting and unexpected findings for diverse user groups: system administrators, HPC system users, and systems researchers (Figure 5). Guidepost provides functionality to select and export data seamlessly back to the notebook context to support a frictionless transition between the exploratory analysis enabled by Guidepost's overview design and confirmatory analysis enabled with programming workflows. In expert evaluations, we found that guidepost promoted the generation of hypotheses about causal relationships in the data among participants. Guidepost is being actively maintained and new features are regularly being added. It is currently deployed and used by HPC data workers at the National Renewable Energy Laboratory (NREL), Lawrence Livermore National Lab (LLNL) and Argonne National Laboratory (ANL).

4. Statistical Analysis Workflows Support with LLMs and Formal Grammars

The research insights drawn from the tools and models I have developed for multimodal data interaction focus primarily on the divisions of responsibilities between visualization and code that make the development and use of complex embedded visualization designs tractable. However, a notable support gap is highlighted by my work: *What are users supposed to do, practically, with selected data after making a discovery with a multimodal tool*? My work defers this question to the users themselves by suggesting it "be used in existing workflows." However, analytical workflows are typically burdened by ad-hoc, fragile, and highly specific code and visualization[11] suggesting a need for further analytical assistance. Due to the highly individualized nature of the specific analytical workflows that lie downstream of initial dataset exploration, the development of one tool or many tools that can serve this need becomes intractably complex. Analyst

[1] https://github.com/hdc-arizona/roundtrip

assistants powered by Large Language Models (LLM), show promise for supporting users developing analysis code but are currently limited by generative randomness that can inhibit productive analysis (Figure 6). To better understand this tension between the promised power and weaknesses of LLMs in analytical intent translation, **I developed a formal, grammar-guided approach to support accurate translation of hypotheses and developed a methodology for measuring where and when an LLM fails to capture intent.**

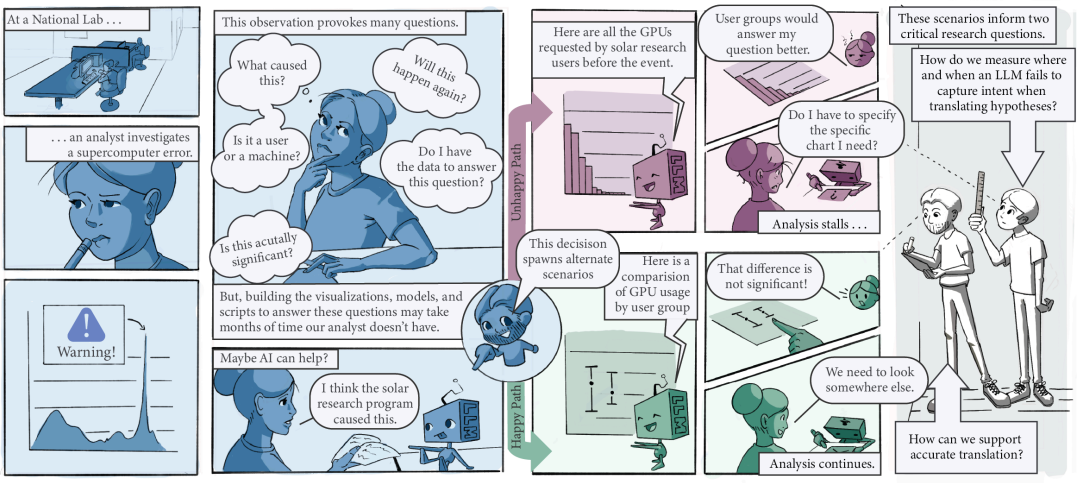


Figure 6: Comic highlighting the dangers and benefits of naive LLM use in analytical workflows in the HPC domain.

Using this formal grammar and a measurement methodology adapted from software testing which decomposes an analytical statement into a set of semantic invariants that should remain the same across representations – I was able to probe the effectiveness of my grammar for generating visualizations from a hypothesis. Initial results from a comparative study – where my grammar was used by an LLM to build an intermediate representation (IR) of an analytical statement before generating a visualization – showed promise in increasing the fidelity and quality of the final visualization compared to a basic prompt, but was not as effective as anticipated. In a follow on study I added comment syntax in this grammar and observed both a measurable increase the accuracy of the generated IR and subsequent visualization while the LLM provided comments that captured the "thinking" which explained their translation process. Notably, it captured why some semantics drifted in the IR (and subsequent visualization), e.g. the LLM specified a Kruskal-Wallis test instead of a direct omnibus comparison of medians because it was more robust, even though it was not explicitly asked for.

This work is currently under review at ACM CHI and is being folded into an extension to my **Guidepost** library called **Mentor**. Although the results from this work have been promising, they have also generated many open-ended questions which I am to explore with **Mentor**. How does the effectiveness of this approach scale to open source or less cutting-edge models? What further interventions are required to get an LLM to faithfully and consistently relay its internal model of a research question? What interfaces and designs can help users align models with an LLM in the midst of already complex and diffuse federated workflows?

5. Future Research Plans

My current and past research highlights a common trend in how software is designed – or not designed – for data workers: little thought is given to the "interstellar space" between software platforms that make up federated workflows. Currently, many new data analysis tools are crafted with care by researchers and designers but exist only as lonely stations with robust dedicated local infrastructure but very few ports to pass information in and out, leading to quick abandonment. The analyst's trip out from one software to the other is presently a perilous journey with no guarantee that they will arrive at the next stop: risking abandoning their task in an "asteroid field" of emails, for example, or losing relevant "cargo" with an accidental call to a copy command or misplaced data dump. This poses a concrete challenge: **How can we enable data workers to move fluidly across tools while preserving analytical state, intent, and understanding?** This challenge is comprised of three constituent challenges that exist at different levels of abstraction: technical, cognitive, and cultural.

[Technical] Preserving analytical state across tools: The primary technical challenge of this issue lies in unpacking how heterogeneous tools can exchange and coordinate data across interfaces so that data workers can work continuously and smoothly across their federated workflows. For example, an annotation made in a domain-specific visualization software should also appear in real time in an excel spreadsheet that has the same data loaded. My prior work on `guidepost` and `%cct` has explored models of data synchronization and fluid movement between data analysis contexts

in depth and highlighted the need for better support for transitions out of one analysis context and into another. I have already begun work on this challenge with my **Mentor** extension to **Guidepost**. I am treating this as a focused design probe that explores the feasibility of integrating an LLM powered analysis assistant across a broad set of analysis software commonly used by data workers in the HPC domain, e.g a centrally hosted Tableau dashboard and a personal computational notebook or command line interface. This work will help to define the tractable engineering problems associated with creating and maintaining a synchronous, shared data model across many software platforms while surfacing critical design considerations – e.g. when synchronicity is desired or not desired – with feedback from real users in real deployment contexts.

[Cognitive] Preserving meaning across representations: The cognitive challenge of supporting analytical continuity across analytical tools lies largely in understanding and spanning the cognitive gulf which remains between data models reified for use in analysis tools and the mental models of data that users hold. When supporting transitions across stable points of software in a federated workflow, a user's understanding of a data model must be crisp in their minds and provide them with a stable cognitive foundation for reasoning about their data and mapping findings made from one interface into the context of another. Tool design should support users in maintaining this crisp model. I explored this issue with a focused inquiry into the mental data models of data workers in HPC performance analysis and found significant room for deeper inquiry that explores the "firmness" and "flexibility" of mental data models across different contexts (e.g. when initially exploring a dataset vs. asking focused analytical questions about that dataset or when using a visualization software vs a code-based interface to data). The formative research into this can continue in the vein of my prior work[5] and related work[12] by soliciting data mental models through purely qualitative elicitation techniques like sketching and interviewing but expanded to data workers in diverse domains to extract more general themes. Verification of identified themes can be driven synergistically by the practical evaluations of software developed to address the technical layer of the greater challenge I investigate.

[Cultural] Integrating new capabilities into established practices: The use of specific analytical tools and workflows can be deeply baked into the culture of some data workers leaving many workers happily siloed in sufficiently usable but underpowered data analytics workflows. I observed occurrences of this among data workers in the HPC domain when interviewing experts to validate my grammar design. Systems administrators in particular made strong socio-indexical identifications that related their common use of command line tools and text editors like "Vim" to their identity as system administrators. These associations make the introduction of new tools into federated workflows difficult and has significant consequences on the size of the design space allowable for interactive data exploration tools. Furthermore, many data formats and tools used are often established by convention and entrenched by decades of exposure across generations of users. The successful adoption of new data analysis tools and representational data bridges between tools hinges on overcoming cultural barriers as much as cognitive and technical ones. To investigate cultural challenges to analysis tool use and adoption I would engage in a foundational set of interviews focused on extracting socio-indexical attitudes among specific subgroups of data workers in the HPC space. Using derived themes I could develop general guidance for deployment and adoption among specific groups of data workers and measure the effectiveness of this guidance on a long timeline with my currently deployed and planned software.

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